

Project Report: Role-Based Access Control Service Management System

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Introduction

This project is a web-based Service Management System with Role-Based Access Control (RBAC), enabling secure user management, service handling, and automation. Built using **Flask** for the backend and **Vue.js** for the frontend, it ensures efficient operations for Admins, Professionals, and Customers.

Key Features

- **RBAC:** Secure access for Admins, Professionals, and Customers with JWT-based authentication.
 - **Admin Dashboard:** Manage users, services, and search/filter options.
 - **Service Requests:** Customers can create/edit requests; professionals can accept/close them.
 - **Automation:** Daily reminders and monthly reports via email or chat.
 - **Caching:** Redis for faster API performance.
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Tech Stack

- **Frontend:** Vue.js
 - **Backend:** Flask
 - **Database:** Flask-SQLAlchemy
 - **Caching:** Redis
 - **Task Queue:** Celery
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Core Models

1. **User:** Manages user details (role, verification, blocking).
 2. **Service:** Handles service information (name, price, time required).
 3. **ServiceRequest:** Tracks service requests with status and remarks.
 4. **Review:** Stores customer reviews and ratings for services.
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Project Structure

- **Backend:** Contains routes, models, tasks, and configurations.
 - **Frontend:** Organized into components, views, and services for scalability.
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[Video](#)